Information Theory & Error Correcting Codes

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Disclaimer

This document contains the lectures given by Dr. Seffah.

To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

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Chapter 0

Remainders

1. Congruences & $\mathbb{Z}/m\mathbb{Z}$ Arithmetic

Definition 1.1 (Congruence Of Integers / Congruence Class): *Let $a, b, m \in \mathbb{Z}$ with $m > 0$, we say that a is congruent to b modulo m and we write $a \equiv b \bmod m$ if $m \mid a - b$ which gives an equivalence relation. The class of a in the congruence relation by m is called the congruence class of a modulo m, which is $\overline{a} = a + m\mathbb{Z}$.*

Theorem 1.2: *Let $a, b, c, d \in \mathbb{Z}$ and $m \in \mathbb{N}$, with $a \equiv b \bmod m$ and $c \equiv d \bmod m$ then we have the following statements are true:*

1. $a + c \equiv b + d \bmod m$.
2. $a - c \equiv b - d \bmod m$.
3. $ac \equiv bd \bmod m$.

We denote $\mathbb{Z}/m\mathbb{Z}$, the set of all congruence classes modulo m.

2. Euler φ Function

Definition 2.3 (φ Function): *The Euler φ function is defined as $\varphi : \mathbb{Z} \to \mathbb{N}$, where $\varphi(n) = \#\{x \in [\![0, m-1]\!] \mid \gcd(x, m) = 1\}$.*

Lemma: *Let $r, m, p \in \mathbb{N}$ where p is prime, then $\gcd(m, p^r) \neq 1 \Leftrightarrow p \mid m$*

Proof. $\Rightarrow$ let $d = \gcd(m, p^r)$, since $d \mid p^r$ and p is prime then $d = p^k$ for some $k \in \mathbb{N}$, and $p^k = d \mid m$ thus $p \mid p^k$ and $p^k \mid m$ so $p \mid m$. $\Leftarrow$ suppose that $p \mid m$ and since $p \mid p^r$ too then $p \mid \gcd(m, p^r)$ given $p > 1$ then $\gcd(m, p^r) \neq 1$. $\square$

Proposition 2.4: *Let p be a prime number, then $\varphi(p) = p - 1$ and $\varphi(p^r) = p^r - p^{r-1} = p^r\left(1 - \frac{1}{p}\right)$.*

Proof. Let $p, r \in \mathbb{N}$ with p prime. From the lemma, we have that for $m \in \mathbb{N}$, $\gcd(p^r, m) = 1 \Leftrightarrow p \nmid m$, the numbers that are divisible by p in $[\![0, p^r - 1]\!]$ are of the form kp where $k \in [\![0, p^{r-1} - 1]\!]$, so the number of numbers that are divisible by p in $[\![0, p^r - 1]\!]$ is $p^r - p^{r-1}$ so $\varphi(p^r) = p^r - p^{r-1}$. $\square$

Theorem 2.5 (Lagrange): *Let G be a finite group and H a subgroup of G, then $\# H \mid \# G$.*

This theorem is already done multiple times in algebra so no need to prove it here.

Lemma 2.6: *Let $n \in \mathbb{N}$, $x \in (\mathbb{Z}/n\mathbb{Z})^*$ (invertible) $\Leftrightarrow \gcd(x, n) = 1$.*

Proof. $\Rightarrow$ Suppose that $x \in (\mathbb{Z}/n\mathbb{Z})^*$, then $\exists y \in \mathbb{Z}/n\mathbb{Z}$ such that $xy = 1 \bmod n$, then $xy = 1 + kn \Rightarrow xy - kn = 1$ by Bezout theorem we have $\gcd(x, n) = 1$. $\Leftarrow$ SUppose that $\gcd(x, n) = 1$, by Bezout theorem, $\exists y, k \in \mathbb{Z}, xy + kn = 1$ and thus $xy + kn = 1 \bmod n \Rightarrow xy = 1 \bmod n$ so $x \in (\mathbb{Z}/n\mathbb{Z})^*$. $\square$

We need this lemma since $\varphi(n) = \#(\mathbb{Z}/n\mathbb{Z})^*$.

Theorem 2.7 (Euler): *Let m be a positive integer modulo a be an integer relatively prime to m then $a^{\varphi(m)} \equiv 1 \bmod m$.*

Proof. Let a relatively prime to m, then $\gcd(a, m) = 1 \Rightarrow a \in (\mathbb{Z}/m\mathbb{Z})^*$, a has finite order thus $\exists k \in \mathbb{N}, a^k = 1 \bmod m$, $\langle a \rangle$ is a subgroup of $(\mathbb{Z}/m\mathbb{Z})^*$ so $k \mid \#(\mathbb{Z}/m\mathbb{Z})^* = \varphi(m)$ and thus $\varphi(m) = kb$, thus $a^{\varphi(m)} = a^{kb} = \left(a^k\right)^b = 1^b = 1 \bmod m$, which completes the proof. $\square$

Corollary 2.8 (Fermat): *Let p be a prime, if the integer a is not divisible by p then $a^p \equiv a \bmod p$.*

Proof. Just take $m = p$ in Euler's theorem, then $\varphi(m) = p - 1$ so $a^{\varphi(m)} = a^{p-1} = 1 \bmod m$, so $a^p = a \bmod m$. □

3. Quadratic Residues

Definition 3.9 (Quadratic Residue): *Let p be an odd prime and a an integer not divisible by p, we say that a is a quadratic residue modulo p if there exists $x \in \mathbb{Z}$, such that $x^2 \equiv a \bmod p$.*

Theorem 3.10: *An integer a is a quadratic residue modulo p if and only if $\gcd(a, p) = 1$ and a has a square rest modulo p.*

Proof. $\Rightarrow$ Suppose that a is a quadratic residue modulo p, that is, there exists $x \in \mathbb{Z}$ such that $x^2 \equiv a \bmod p$. If $\gcd(a, p) \neq 1$ then $p \mid a$ so $a \equiv 0 \bmod p$ thus $x = 0$, so necessarily $\gcd(a, p) = 1$ and a has x^2 remainder when divided by p thus the rest is a perfect square. $\Leftarrow$ Suppose now that $\gcd(a, p) = 1$ and a has a square rest modulo p, then $a \not\equiv 0 \bmod p$ and $a \equiv x^2 \bmod p$ thus a is quadratic residue. □

Definition 3.11 (Legendre Symbol): *Let p be an odd prime, define the Legendre symbol as:*

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } \gcd(a, p) = 1 \text{ and } a \text{ is a quadratic residue} \bmod p \\ -1 & \text{if } \gcd(a, p) = 1 \text{ and } a \text{ is not a quadratic residue} \\ 0 & \text{if } p \text{ divides } a \end{cases}$$

Theorem 3.12: *Let p be an odd prime, for every integer a*

$$\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \bmod p$$

Proof. We treat by cases

1. $p \mid a$: then $a = 0 \bmod p \Rightarrow a^{\frac{p-1}{2}} = 0^{\frac{p-1}{2}} = 0 \bmod p$.
2. $p \nmid a$: then $\gcd(a, p) = 1$, by Fermat's little theorem we have that $a^{p-1} \equiv 1 \bmod p$, the square roots of 1 are -1 and 1 only thus $a^{\frac{p-1}{2}} \equiv 1$ or $-1 \bmod p$.
   1. If a is a quadratic residue then $\exists x \in \mathbb{Z}, x^2 \equiv a \bmod p \Rightarrow a^{\frac{p-1}{2}} \equiv x^{2\frac{p-1}{2}} \equiv x^{p-1} \equiv 1 \bmod p$.
   2. If $a^{\frac{p-1}{2}} \equiv 1 \bmod p$, we have that $\mathbb{Z}_p$ is a finite field thus there exists a generator g of $(\mathbb{Z}_p)^*$ (primitive element theorem in finite fields), then $a \equiv g^k \bmod p$, we have then $a^{\frac{p-1}{2}} \equiv g^{k\frac{p-1}{2}} \equiv 1 \bmod p$. Given that the generator g has order $p - 1$ then $k\frac{p-1}{2} \equiv 0 \bmod p - 1$ thus $k\frac{p-1}{2} = m(p-1) \Rightarrow k = 2m$ so k is even, thus $a \equiv g^{2m} = (g^m)^2 \bmod p$ so a is a quadratic residue modulo p.

Given that a is a quadratic residue modulo p if and only if $a^{\frac{p-1}{2}} \equiv 1 \bmod p$ and $a^{\frac{p-1}{2}} \equiv 1$ or $-1 \bmod p$ then necessarily if a is not a quadratic residue, $a \equiv -1 \bmod p$. □

Exercise 3.13:

1. *Decompose into partial fractions in $\mathbb{R}[x]$ the rational function*
$$\frac{x}{x^4 + x^2 + 1}$$
2. *Let K be a commutative field, and let $p : x^2 + \lambda x + \mu$ be a monic polynomial of degree 2, show that p is reducible over K if and only if it has a root in K.*
3. *Let $K = \mathbb{Z}/5\mathbb{Z}$ be the field of residue classes, factor the polynomial $(x^2 + 4)(x^2 + 3)$ into irreducible factors over K.*
4. *Still with $K = \mathbb{Z}/5\mathbb{Z}$, decompose into partial fractions the rational function*
$$\frac{x - \overline{2}}{(x^2 + \overline{4})(x^2 + \overline{3})}$$

1. Decomposing into partial fractions: we have that the denominator decomposes into the following two irreducible polynomials

$$\begin{aligned} x^4+x^2+1 &= x^4+2x^2+1-x^2 \\ &= (x^2+1)^2-x^2 \\ &= (x^2+x+1)(x^2-x+1) \end{aligned}$$

and thus we have that the fraction decomposes as follows

$$\frac{x}{x^4+x^2+1} = \frac{ax+b}{x^2+x+1} + \frac{cx+d}{x^2-x+1}$$

by multiplying both sides by x^4+x^2+1 we get

$$x = (ax+b)(x^2-x+1)+(cx+d)(x^2+x+1)$$

$$0 = ax^3-ax^2+ax+bx^2-bx+b+cx^3+cx^2+cx+dx^2+dx+d-x$$

$$0 = (a+c)x^3+(-a+b+c+d)x^2+(a-b+c+d-1)x+(b+d)$$

$$\Rightarrow \begin{cases} a+c &= 0 \\ -a+b+c+d &= 0 \\ a-b+c+d-1 &= 0 \\ b+d &= 0 \end{cases} \Rightarrow \begin{cases} a=0 \\ b=-1/2 \\ c=0 \\ d=1/2 \end{cases}$$

thus we get

$$\frac{x}{x^4+x^2+1} = \frac{1}{2(x^2-x+1)} - \frac{1}{2(x^2+x+1)}$$

2. Let K be a commutative field and let $P(x)=x^2+\lambda x+\mu$. $\Rightarrow$ Suppose that $P(x)$ is reducible, then $P(x)=Q(x)\cdot R(x)$ with $\deg(Q)\neq 0$ and $\deg(R)\neq 0$ and since $\deg(Q)+\deg(R)=\deg(P)$ then necessarily $\deg(Q)=\deg(R)=1$, so $Q(x)=ax+b$ and $R(x)=cx+d$ and since $a,b\in K$ then $-\frac{a}{b}\in K$ which is a root of Q and thus a root of P so P has a root in K. $\Leftarrow$ Suppose that $P(x)$ has a root α, then $(x-\alpha)$ divides P and thus we get that P is reducible.
3. Let $K=\mathbb{Z}/5\mathbb{Z}$, factor $(x^2+4)(x^2+3)$ into irreducible factors over K, $-3=2 \bmod 5$ is not a quadratic residue since $2^{\frac{5-1}{2}}=4=-1 \bmod 5$ thus x^2+3 is irreducible given it has no roots. While $-4=1$ is a quadratic residue with roots -1 and 1 so $(x^2+4)(x^2+3)=(x^2-1)(x^2+3)=(x-1)(x+1)(x^2+3)$.
4. Decomposing now the partial fraction in $\mathbb{Z}/5\mathbb{Z}$,

$$\begin{aligned} \frac{x-2}{(x^2+4)(x^2+3)} &= \frac{x-2}{(x+4)(x+1)(x^2+3)} \\ &= \frac{a}{x+4}+\frac{b}{x+1}+\frac{cx+d}{x^2+3} \end{aligned}$$

by multiplying all parts by $(x^2+4)(x^2+3)$ we get

$$x-2 = a(x+1)(x^2+3)+b(x+4)(x^2+3)+(cx+d)(x+4)(x+1)$$

we set the following values for x

- $x=-1$: $-3=b(-1+4)(1+3)=12b=2b \Rightarrow b=3*(-3)=-9=1$.
- $x=+1$: $-1=a(1+1)(1+3)=8a=3a \Rightarrow a=2*(-1)=-2=3$.

Exercise 3.14:

1. *Decompose* 561 *into prime factors.*
2. *Let* a *be an indeterminate, for any integer* $n\geq 1$*, justify the formula*
$$a^n-1=(a-1)(a^{n-1}+a^{n-2}+\cdots+a+1)$$
3. *Justify for any integer* $m\geq 1$ *and any integer* $n\geq 1$ *the formula*
$$a^{mn}-1=(a^m-1)(a^{(n-1)m}+a^{(n-2)m}+\cdots+a^m+1)$$
4. *Let* $a\in\mathbb{Z}$ *be an integer, prove* $a^{561}-a=a(a^{2\cdot 280}-1)=a(a^2-1)m$ *for some integer* m.
5. *For any* $a\in\mathbb{Z}$*, show that* $a(a-1)(a+1)$ *is divisible by* 3.
6. *Show that* $a^{561}-a$ *is a multiple of* 3.
7. *Show that* $a^{561}-a$ *is a multiple of* 17.
8. *Show that* $a^{561}-a$ *is a multiple of* 11.
9. *Show that* $a^{561}\equiv a \bmod 561$.

1. The prime decomposition of 561 is $3\cdot 11\cdot 17$.
2. Let a be a formal variable, we have
$$(a-1)\sum_{i=0}^{n-1}a^i = \sum_{i=0}^{n-1}a^{i+1}-a^i = \sum_{i=1}^{n}a^i-\sum_{i=0}^{n-1}a^i = a^n-1$$
3. By taking $a=a^m$ in the previous inequality we get the result.

4. Clearly $561 = 2 \cdot 280 + 1$ thus $a^{561} - a = a(a^{560} - 1) = a(a^{2 \cdot 280} - 1)$ and we have from the previous question by taking $m = 2$ and $n = 280$ that $a^{561} - a = a(a^2 - 1)\left(\sum_{i=1}^{279} a^{2i}\right) = a(a^2 - 1)m$.
5. Let $a \in \mathbb{Z}$, we have that $a = 3q + r$ with $0 \leq r < 2$ from the Euclidean division algorithm, then $a(a-1)(a+1) = (3q + r)(3q + (r-1))(3(q+1) + (r-2))$ and since $0 \leq r < 2$ then necessarily either r, $r - 1$ or $r - 2$ is 0, and thus it has one of the factors divisible by 3.
6. $a^{561} - a$ is divisible by 3 since $a^{561} - a = a(a+1)(a-1)m$ which is divisible by 3 from the previous question.
7. $a^{16} = 1 \bmod 17$ by Fermat little theorem, then $a^{560} = a^{16 \cdot 35} = 1 \bmod 17$ thus, $a^{561} = a \bmod 17 \Rightarrow 17 \mid a^{561} - a$.
8. Same here, $a^{10} = 1 \bmod 11$ by Fermat little theorem, then $a^{560} = a^{10 \cdot 56} = 1 \bmod 11$ thus, $a^{561} = a \bmod 11 \Rightarrow 11 \mid a^{561} - a$.
9. Given that all the divisors of 561 divide $a^{561} - a$ then it is divisible by their product so $561 \mid a^{561} - a \Rightarrow a^{561} \equiv a \bmod 561$.

Chapter 1

Introduction To Information Theory

The digital revolution has transformed how we communicate, store and process information. This chapter introduces the fundamental concepts of information theory, a discipline at the intersection of mathematics, computer science and telecommunication.

This diagram present the general structure of any communication system. The communication process begins with an information source. We start by thinking what a message could be. In general, we can consider a message to be constructed by a set of symbols $\mathcal{X} = \{x_1, ..., x_n\}$ and given a context, has a distribution of which of those symbols are sent. Thus, we consider a message consecutive realizations of a random variable X defined in $\mathcal{X}$.

Now the goal is to quantify information, thus given $X \sim P_X(x)$, we define the amount of information for some symbol x to be $I(x) = -\log_2(P_X(x))$ (justifications for these definitions will be in the next chapter). Then, we can define an average amount of information from a source X, which is called the entropy $H(X) = \mathbb{E}[I(X)]$. Such a quantity is important given Shannon's first theorem, which gives a lower bound for any data compression.

After being able to measure the amount of information, we can measure efficiency of some encodings both for source and channel by how much redundancy is added to how much is actually needed.

1.1. The Communication System Model

Shannon's groundbreaking contribution was a formal model that describes communication between two entities, whether machine or humans. This model highlights the essential components:

- Source: the originator of the information.
- Channel: The medium through which information is transmitted.
- Recipient: The receiver of the information.

1.1.1. Source & Message

The sources generate a message composed of symbols from a specific alphabet.

- Discrete data: text, numbers, symbols.
- Digitized analog data: image, voice, video converted to digital form.

1.1.2. Transmitter Processing

The transmitter performs three crucial operations.

1. Source Coding: data compression to minimize message size.
2. Channel Coding: adding redundancy for error correction.
3. Modulation: converting digital messages to physical signal.

1.1.3. Receiver Processing

The receiver performs these receiver operations.

1. Demodulation: converting physical signal into a digital message.
2. Channel Decoding: error detection and correction.
3. Source Decoding: reconstruction of the original message.

Chapter 2

Information & Entropy

Information theory is a theory founded by Claude Shannon, providing a mathematical framework for quantifying information, studying its transmission, processing, and storage. This chapter focuses on the fundamental concepts of information measurement through entropy and related measures.

2.1. Quantifying Information

Notation: *Let Σ be a set:*

- $\Sigma^n = \Sigma \times \cdots \times \Sigma$ *is the set of words of length n. We take that $\Sigma^0 = \{\varepsilon\}$ where ε is the empty word.*
- $\Sigma^* = \cup_{i \geq 0} \Sigma^n$ *the set of all possible words over Σ.*
- $\Sigma^+ = \cup_{i \geq 1} \Sigma^n$ *the set of all possible non-empty words over Σ.*
- *For $x \in \Sigma^*$, we define $|x| = \min\{n \in \mathbb{N} \mid x \in \Sigma^n\}$.*
- *The set of binary letters as $\mathbb{B} = \{0, 1\}$.*
- *For $x = x_1 ... x_n, y = y_1 ... y_m \in \Sigma^*, xy = x_1 ... x_n y_1 ... y_m \in \Sigma^*$.*

Definition 1 (Symbol/Message): *Let $\mathcal{X} = \{x_1, ..., x_m\}$ be a finite set of alphabets. An element of $\mathcal{X}$ is called a symbol, and elements of $\mathcal{X}^+$ are called a message with alphabet $\mathcal{X}$.*

To model information that was given from a message, we consider the fact that a message contains more information the more unpredictable it is. For example, consider a sender that keeps sending the message "Hello!" over and over again, at each iteration, it becomes predictable what the next message would be, thus there is no more information that is passed. Uncertainty is a property of a random process, thus a good way to model a message is to consider it as a random variable X that has values in $\mathcal{X}$. From the previous example too, we can think that the information in a message is related to how uncertain it is, that is, how less probable it would be sent. Also, notice that the information from independent events should be additive, that is, if even p and q happen that are independent, then the amount of information gained from the even pq happening is the sum of information gained from both the events p and q.

Notation: *For the sake of simplifying notation, we will denote $P(X = x)$ as $P(x)$, same for other random variables, like $P(Y = y)$ as $P(y)$. Same for all the remaining notations like $P(y|x)$ which is $P(Y = y|X = x)$.*

To define a measure for information $I : \mathcal{X} \to \mathbb{R}$, we start by considering some random variable X that represents a source of symbols $\mathcal{X}$. As discussed before, the information will be a function of probability, thus we take $I(x) = \varphi(P(x))$ from our previous discussion it would satisfy the following postulates:

- Postulate 1: Information depends only on probability $I(A) = \varphi(P(A))$.
- Postulate 2: Certain events provide no information $\varphi(1) = 0$.
- Postulate 3: Information decreases with increasing probability, φ is decreasing.
- Postulate 4: Impossible events provide infinite information $P(0) = \infty$.
- Postulate 5: Information from independent events $\varphi(pq) = \varphi(p) + \varphi(q)$.

Also, clearly φ would be continuous since a tiny difference in probability would result in a tiny difference in the amount of information that it would result in.

We will briefly prove that $\varphi \equiv -\log_b$ for any base b. By induction we have that $\varphi(a^n) = nf(a)$ and since $\varphi(a) = \varphi\left(a^{q/q}\right) = q\varphi(a^q)$ then necessarily $\varphi\left(a^{p/q}\right) = (p/q)\varphi(a)$. By continuity, we have $\forall x \in \mathbb{R}, \varphi(a^x) = x\varphi(a)$. Now take $t = \log_a x$, then $\varphi(x) = \varphi(a^t) = t\varphi(a) = \varphi(a)\log_a x$, thus $\varphi(x) = \varphi(a)\log_a(x)$, by taking $x \to 0$, we have that $\varphi(a)\log_a(x) \to +\infty$ so $\varphi(a) < 0$. Therefore, $\varphi \equiv -\log_b$ for some b.

Definition 2 (Measure Of Information): *Let X be a random variable that represents the probability that a message x is sent as $P(X = x) = P(x)$, we define the measure of information of the message x as*

$$I(x) = -\log_2 P(x)$$

$\log_2$ is taken for the usual reason that information is represented in binary, and that if it is represented in any other base, it would have just a linear factor added. In practice, transmission lines tend to cause irregularities in the signal, that is, it alters the contents of the message with some noise. Thus, in transmission lines we are interested in the distribution of the message y that are received, given some sent message x, which will be measured by $P(y|x)$.

2.2. Entropy

After we defined a measure of information, which we discussed to be a measure of uncertainty, we need a way to quantify the average amount of uncertainty for all the values of X. Thus, we define the entropy as follows.

Definition 3 (Entropy): *Let X be a random variable with values in $\mathcal{X}$, we define the entropy $H(X)$ as the average amount of information, that is*

$$H(X) = \mathbb{E}[I(X)] = -\sum_{x \in X} P(x) \cdot \log_2 P(x)$$

Notice that the information of some message x depends only on its unpredictability, that is, its probability of occurrence $P(x)$ not its value.

Proposition 2.2.4: *Let X, Y with symbols in $\mathcal{X}, \mathcal{Y}$ be two i.i.d sources.*

1. *Non-negativity: $H(X) \geq 0$.*
2. *Maximum Entropy: $H(X) \leq \log_2 \# \mathcal{X}$.*
3. *Independent Additivity: $X \perp Y \Rightarrow H(XY) = H(X) + H(Y)$.*

Proof.

1. $\forall x \in \mathcal{X}, I(x) \geq 0$, then $H(X) = \mathbb{E}[I(X)] \geq 0$
2. Here, we optimize on the distribution of X, $\max_P H(X) = -\min_P \sum_{x \in X} P(x) \cdot \log_2 P(x)$, hence for I a finite set $\min_{\{y_i \in [0,1]\}_{i \in I}} \sum_{i \in I} y_i \log_2(y_i)$ with $\sum_{i \in I} y_i = 1$. Notice that all the terms have the same sign so $\min \sum_{i \in I} \ldots = \sum_{i \in I} \min \ldots$. y_i all having the same range would mean that we just need to find c such that $y_i = c$ and satisfies the second inequality, which is clearly $c = \frac{1}{\# I}$.

So X would be a uniform distribution over $\mathcal{X}$, $H_u(X) = \mathbb{E}[I(X)] = -\sum_{x \in X} 1/(\# \mathcal{X}) \log_2(1/(\# \mathcal{X})) = \log_2 \# \mathcal{X} \geq H(X)$.

1. Let X, Y be two independent variables, then $H(XY) = \mathbb{E}[I(XY)] = \mathbb{E}[I(X) + I(Y)] = \mathbb{E}[I(X)] + \mathbb{E}[I(Y)] = H(X) + H(Y)$. □

Example: Let $\mathcal{X} = \{0, 1\}$, and X a discrete random variable with values in $\mathcal{X}$, which has the distribution

$$P(x) = \begin{cases} 0.9 \text{ if } x = 0 \\ 0.1 \text{ if } x = 1 \end{cases} \quad \Rightarrow \quad I(x) = \begin{cases} 0.152 \text{ if } x = 0 \\ 3.321 \text{ if } x = 1 \end{cases}$$

And the measure of average amount of information is $H(X) = 0.4689$.

Example: DNA sequences provide an excellent example of entropy application. Let's consider:

- Primate DNA: $P(A) = P(G) = P(C) = P(T) = 1/4$. $H(X) = 2$
- Bacterial DNA: $P(G) = P(C) = 0.38$, $P(A) = P(T) = 0.12$. $H(X) = 1.795$.

The entropy of bacterial DNA indicates more predictable sequences compared to the primate DNA.

2.2.1. Joint Entropy

Definition (Joint Entropy): *Let X, Y be two random variables with joint distribution $P(x, y)$, the joint entropy is*

$$H(X, Y) = -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(x, y)$$

Proposition 2.2.1.5: *Let X, Y be two i.i.d sources.*

1. *Symmetry: $H(X, Y) = H(Y, X)$.*
2. *Non-Negativity: $H(X, Y) \geq 0$.*
3. *Upperbound: $H(X, Y) \leq H(X) + H(Y)$.*
4. *Lowerbound: $H(X, Y) \geq \max(H(X), H(Y))$.*

Proof.

1. trivial by $P(x, y) = P(X = x, Y = y) = P(Y = y, X = x) = P(y, x)$.

2. $\forall x \in X, \forall y \in Y, 0 \leq P(x,y) \leq 1 \Rightarrow \log_2 P(x,y) \leq 0$ so we have $P(x,y)\log_2 P(x,y) \leq 0$. Hence,

$$H(X,Y) = -\sum_{x\in X}\sum_{y\in Y} P(x,y)\log_2 P(x,y) \geq 0$$

3. We have $H(X,Y) - H(X) - H(Y) = \sum_{x\in X} P(x)\log_2 P(x) + \sum_{y\in Y} P(y)\log_2 P(y) - \sum_{(x,y)\in X\times Y} P(x,y)\log_2 P(x,y)$, by using the fact that $P(x) = \sum_{y\in Y} P(x,y)$, we can transform all the sums to be indexed with $X \times Y$ as follows:

$$\sum_{x\in X} P(x)\log_2 P(x) = \sum_{(x,y)\in X\times Y} P(x,y)\log_2 P(x)$$
$$\sum_{y\in Y} P(x)\log_2 P(y) = \sum_{(x,y)\in X\times Y} P(x,y)\log_2 P(y)$$

Thus we obtain the writing

$$H(X,Y) - H(X) - H(Y) = \sum_{(x,y)\in X\times Y} P(x,y)\log_2\left(\frac{P(x)P(y)}{P(x,y)}\right)$$

We have that $\forall t \geq 0, \log_2(t) \leq t - 1$ thus

$$\begin{aligned}
H(X,Y) - H(X) - H(Y) &= \sum_{(x,y)\in X\times Y} P(x,y)\log_2\left(\frac{P(x)P(y)}{P(x,y)}\right) \\
&\leq \sum_{(x,y)\in X\times Y} P(x,y)\left(\frac{P(x)P(y)}{P(x,y)} - 1\right) \\
&\leq \sum_{(x,y)\in X\times Y} P(x)P(y) - P(x,y) \\
&\leq \sum_{(x,y)\in X\times Y} P(x)P(y) - \sum_{(x,y)\in X\times Y} P(x,y) \\
&\leq 1 - 1 = 0
\end{aligned}$$

Therefore

$$H(X,Y) \leq H(X) + H(Y)$$

4. Without loss of generality, we prove that $H(X,Y) \geq H(X)$, we have from before that $P(x) = \sum_{y\in Y} P(x,y)$ and since $\forall (x,y) \in X \times Y, P(x,y) \geq 0$ then $\forall (x,y) \in X \times Y, P(x,y) \leq P(x)$.

$$\begin{aligned}
H(X,Y) &= -\sum_{(x,y)\in X\times Y} P(x,y)\log_2 P(x,y) \\
&\geq -\sum_{(x,y)\in X\times Y} P(x)\log_2 P(x) = H(X)
\end{aligned}$$

and same to prove that $H(X,Y) \geq H(Y)$ and hence we obtain the inequality $H(X,Y) \geq \max(H(X), H(Y))$. $\square$